Subject: Embedded System Lab

Subject Code: 7IT06

Experiment No. 1

Title: Demonstration of Arduino Uno Development Board, Various Sensors, and Installation of Arduino IDE Software.

Objectives:

1. To understand the hardware architecture of the Arduino Uno development board.
2. To install the Arduino IDE and configure it for development.
3. To study and interface basic sensors and modules like:
    - LED (for digital output)
    - DHT11 Temperature and Humidity sensor (for analog/digital input)
    - OLED Display (for data display using I2C communication)

Required Components/Kit:

- Arduino Uno Board
- USB Cable
- LED
- Resistor (220Ω)
- Breadboard
- Jumper Wires
- DHT11 Temperature and Humidity Sensor
- OLED Display Module (128x64, I2C)

Required Software:

- Arduino IDE (Windows/Linux/Mac)
- Drivers for Arduino Uno (installed with IDE)

Procedure:

Step 1: Arduino IDE Installation

1. Visit: https://www.arduino.cc/en/software

2. Download Arduino IDE for your operating system.
3. Install the software and launch Arduino IDE.
4. Connect Arduino Uno via USB and allow driver installation.
5. Select the correct board and port:
    - Go to **Tools > Board > Arduino Uno**
    - Go to **Tools > Port > Select the correct COM port**

Step 2: Understanding Arduino Uno

- ATmega328P microcontroller (main processor)
- 14 Digital I/O pins (0–13)
- 6 Analog input pins (A0–A5)
- USB interface for programming
- Power: 5V via USB or 9V via external supply

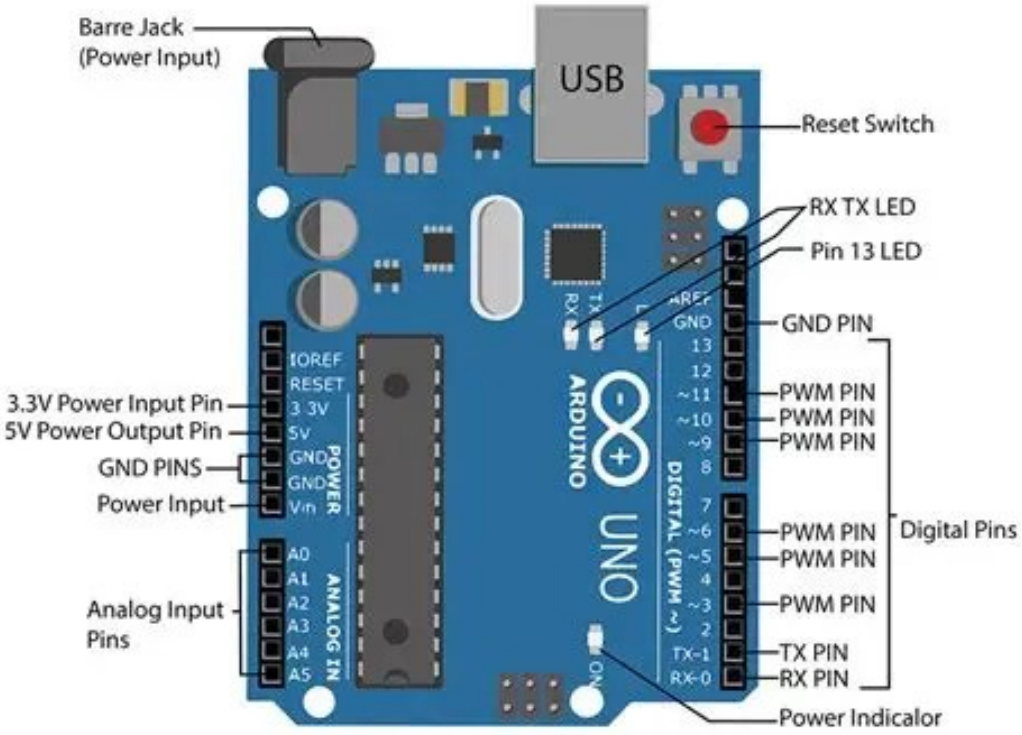


Figure: Hardware Architecture of the Arduino Uno Development Board

Step 3: Interfacing Components

a) LED:

- Connect the long leg (anode) to pin 13 via a 220Ω resistor.

- Connect the short leg (cathode) to GND.

b) DHT11 Sensor:

- VCC to 5V, GND to GND, Data to Digital Pin 2.

c) OLED Display (I2C):

- VCC to 5V, GND to GND, SDA to A4, SCL to A5.

Conclusion:

Students successfully installed Arduino IDE, explored Arduino Uno hardware, and interfaced basic sensors and displays for real-time output.